QGenomix

Quantum-assisted genetic screening for early-onset type 2 diabetes in lean South Asians

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1. The project in one paragraph

South Asians develop type 2 diabetes younger and at a lower body weight than Europeans, so BMI-based screening and India's IDRS questionnaire miss many lean people at risk. QGenomix measures how much genetics adds to clinical screening, tests whether two diabetes mechanisms (insulin production and fat distribution) interact, and asks honestly whether quantum computing helps find the interacting gene pairs. It runs on the real genomes of the 1000 Genomes Project with simulated outcomes (so every discovery can be checked against a known truth), compares 17 predictors in 17 experiments with a classical control for every quantum step, and ends in a working application that turns a consumer DNA file into an explained risk report.

2. What we built

Stage	What we built
Data pipeline	2,504 real genomes (84 M variants) processed with PLINK; the South Asian D-PRISM risk score (1.34 M variants) and five published mechanism scores (62 SNPs); a library of 978 South Asian haplotypes
Cohort and simulator	10,000 synthetic South Asians built from real haplotypes; age, sex, BMI, waist, activity, family history and the IDRS; simulated diabetes with 10 planted gene pairs as known truth
Interaction search	Vectorised score test over all 1,263 cross-mechanism SNP pairs, run inside cross-validation
Pair selection (quantum)	QUBO that rewards relevant pairs and penalises redundant ones, solved by simulated annealing and by QAOA with a constraint-preserving XY mixer; error mitigation for IBM hardware noise
Quantum encoder	8-qubit variational circuit: 12 circuits encode 48 gene pairs into 96 features; our own exact simulator, 7× faster than PennyLane
Models and evaluation	17 predictors (IDRS, BMI rule, PRS, logistic regression, random forest, SVM, XGBoost, quantum variants, stacked hybrid); 17 experiments with DeLong, McNemar and Holm-corrected tests
Application	Dashboard, REST API and one-page PDF report; reads 23andMe and AncestryDNA files and fills in missing variants (LD imputation); Docker image

3. Key results

Question	Result
Does genetics add to clinical screening?	AUC 0.812 → **0.874** with genetics (DeLong $p \le 4\times10^{-6}$); the IDRS alone 0.761, a BMI rule 0.669
Does it find lean people at risk?	Lean diabetes cases caught in the top 20% of risk: BMI rule 0%, IDRS 30%, clinical model 35%, **with genetics 58%**
Does the search find interacting pairs?	**70–84%** of planted pairs in the top 10 at 10,000 people (PLINK BOOST 40–48%); with no true interaction, 18 of 20 runs report nothing ($\lambda = 1.05$)
Does it work on real genomes?	AUC 0.881 on real South Asians; 0.792 on Africans
Is it fair?	Genetics helps in all five South Asian populations and both sexes (AUC 0.84–0.92)
Is it robust to our assumptions?	Genetics beats clinical factors in **13 of 13** simulator settings, and beats the IDRS on lean cases in 13 of 13

Question	Result
Do partial DNA files work?	At 45% coverage, LD imputation cuts the risk-percentile error from 13 to **6** points and keeps 80% of high-risk people flagged (54% before)
Did quantum computing help?	**It does its job, no speed-up**: on gene-pair-only risk, entangled circuit 0.673 vs 0.523 without entanglement; 12-qubit QAOA matches annealing on the full selection. Realistic cohort: all encoders tie (0.866, 0.871, 0.870)
Are the lifestyle what-ifs right?	Waist −5 cm: −28% in the app vs −28% true; exercise −32% vs −40%

4. Progress since the mid-semester evaluation

Approved objective	Mid-sem	Now
Data pipeline	100%	Complete; rebuilt on real 1000 Genomes data
Quantum (VQC) encoder	20%	Complete, with no-entanglement and classical-twin controls
Classical baselines and PRS	15%	Complete, plus the IDRS, a BMI rule and a clinical model
Hybrid stacking ensemble	0%	Complete (best AUC 0.877)
SHAP refinement loop	0%	Built and measured; replaced by the score test + QUBO (73% vs 33% of planted pairs found)
Benchmark and statistical tests	10%	Complete: 17 models, 15 experiments

Added beyond the plan: QAOA with error mitigation; clinical factors and the IDRS; real-genome, fairness and robustness tests; LD imputation for consumer DNA files; the application with PDF report, REST API and Docker.

5. Deliverables

- **Code:** github.com/coderboic/QGenomix — full pipeline, 41 automated tests, continuous integration, Docker; every result reproducible from one configuration file.
- **Report:** QGenomix_Final_Mentor_Evaluation_Report.docx (department format).
- **Slides:** QGenomix_Mentor_Evaluation_Presentation.pptx (41 slides) and QGenomix_Final_Panel_Evaluation_Presentation.pptx (42 slides, with the live demonstration).
- **Application:** dashboard, REST API, PDF report, model card; runs offline on a laptop.

6. Kept for the final panel

- **Real IBM quantum run:** script ready and rehearsed on the Kingston noise model; needs a free IBM account (about 2 of 10 free QPU minutes).
- **Public online demo:** Docker image built and tested; needs deployment under a team account.
- Final report polish, paper draft, demo video and rehearsal.

7. Likely questions and short answers

Your outcomes are simulated. Why trust the results?

Genotypes and risk weights are real; the simulator matches Indian prevalence; the conclusions hold in all 13 alternative settings and on real genomes of five ancestries. Validation on real Indian patients (for example GenomeIndia through IBDC) is the next step.

What did quantum computing actually contribute?

Each part does its job, with no speed-up claimed. On an outcome made only of gene pairs, entanglement is what lets the circuit see the interaction (0.673 with it, 0.523 without), and 12-qubit QAOA makes the full pair selection as well as simulated annealing. On the realistic cohort the quantum models tie their classical controls, and error mitigation recovers most of the quality that hardware noise destroys.

Why a QUBO instead of simply taking the top pairs?

Top-K wastes slots on redundant pairs that share a SNP. The QUBO covers 62 distinct SNPs against 49 and has 26% fewer redundant pairs.

How did you avoid data leakage and overfitting?
An untouched 20% test set; all pair selection and model fitting inside five-fold cross-validation; three seeds; Holm-corrected significance tests; a null experiment with no true interactions.

Why did you drop the SHAP refinement loop?
We built and measured it: SHAP ranking found 33% of planted pairs, the score test 73%. SHAP now explains each person's report instead.

What is new in this work?
Testing interaction between T2D mechanisms for lean South Asians against India's own screening tool; a constraint-preserving QAOA with error mitigation, benchmarked against classical controls; an LD imputation panel for consumer DNA files; and an explainable end-to-end application.

8. Two-minute demo

1. Run `make app` and open localhost:8501/?persona=asha_lean_high&autorun=1.
2. Asha, 34, BMI 21.6: QGenomix says **high risk (84th percentile)**; the IDRS says moderate; the BMI rule does not flag her.
3. Show the mechanism bars: insulin production 96th and fat distribution 88th percentile, both high.
4. Move the what-if sliders: exercise plus a smaller waist cut her risk from 23% to 6%.
5. Press **Download PDF report**. Backup if the laptop fails: screenshots in figures/ or `docker compose up`.